TEAM BRIEFING · 29 OCTOBER 2026

Thibitisha

Let the person who paid tell the owner.

The problem, the idea, how it works, what we have built, and what is left to do. “Thibitisha” is Swahili for “confirm”.

Written for the team on the morning of the Africa’s Talking Open & WIT Hackathon, Nairobi Garage Kilimani. “Thibitisha” is a working name; check it is free before using it publicly.

In one minute

- **The problem:** owners who are not on site cannot see the cash, or the M-Pesa payments sent to a staff member’s own number. They only know what staff tell them.
- **Why it is hard:** every tool so far asks the staff member to record the money. The person who takes money is the one asked to report it.
- **Our idea:** ask the other person in the payment, the one who paid. A tenant, guest or member dials a USSD code and confirms what they paid. We match it with what staff logged. A match sends the payer a receipt. A mismatch alerts the owner.
- **Why we believe it:** governments already use customers to check shops for tax. In Brazil, reported retail revenue rose at least 21%.
- **Where we are:** the USSD engine, staff logging, owner SMS alerts and database are built and tested. Today we add the payer side, the matching, receipts and the Marketplace listing.

1. A story: twelve flats and one caretaker

The people below are made up, but every part of the story comes from what we found in research.

Wanjiru works as a nurse in Doha. With her savings she built twelve flats in Ruiru, outside Nairobi, and pays a caretaker, Otieno, to collect rent and look after the building. Rent is KES 8,000 a month. On paper that is KES 96,000 every month.

Some tenants pay cash. Some send M-Pesa to Otieno's own number because that is who they see. At the end of the month, Otieno sends Wanjiru a WhatsApp message: *"Nine paid. Three are late, I am following up."* She receives KES 72,000.

Wanjiru has no way to check. If she calls the tenants herself, she looks like she does not trust her caretaker, and some of them will not pick up. If she asks everyone to pay into her own till, Otieno says the tenants refuse, and he still needs money for repairs.

Now meet Akinyi, in flat B4. She paid her KES 8,000 in cash on the 28th, as she always does. She did not get a receipt. Two months later, Otieno says she is behind. It is her word against his.

> **Three people lose here.** Wanjiru loses money she cannot see. Akinyi cannot prove she paid. And if Otieno is honest, he has no way to prove *that* either; every short month makes him look like a thief.

The same story repeats with a lodge owner and their night manager, a chama and its treasurer, a milk cooperative and its collection clerk. Someone collects the money; the owner is somewhere else; the payer has no receipt.

2. The problem in plain words

Owners who are not on site cannot see the money that does not reach them. M-Pesa shows an owner every shilling that lands in their own till. It shows nothing about cash, or about payments sent to a caretaker's or manager's number. That is exactly where money goes missing.

Why the usual fixes do not work

Fix people try	Why it fails
"Pay everything to my till."	Payers prefer cash or the person they deal with; staff need cash for float and repairs; rural places run on cash.
A POS or a logbook for staff	It records only what staff choose to enter. The staff member who takes money does not log it.
Cameras and GPS trackers	They cost money, need data, and get switched off. Matatu owners report tracking wires cut within hours of fitting.
Receipt apps	They need a smartphone. Kenya still has 27.4 million feature-phone devices (Communications Authority, June 2026).
Calling payers yourself	Slow, awkward, and it tells your staff you do not trust them.

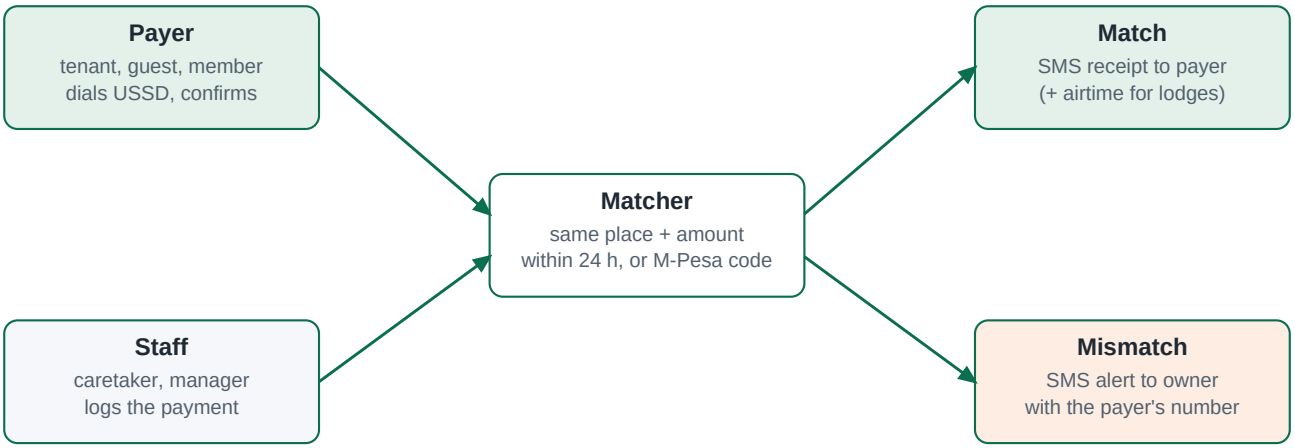
How we found this

We did not start here. On 14 October we built a USSD "Shadow POS" for lodges and bars: the manager logs each room and bar sale, the owner gets SMS alerts, and the cleaner's room count is checked against the manager's. It works, and it is tested. But when we walked through real venues, we kept hitting the same wall: **the manager who steals is the one we are asking to log.** A cleaner can check a lodge, but most small places have no second person.

Research into how others solved this pointed somewhere we had not looked: governments that make the *customer* the checker. That turned the problem around.

3. The idea: let the payer be the auditor

Every payment has two people in it. Until now, only one of them (the staff member) has been asked what happened. Thibitisha asks the other one.



Step	What happens
1. Payer confirms	The tenant, guest or member dials a USSD code on any phone, enters the place code and the amount, and confirms. It takes under a minute.
2. Staff log	The caretaker or manager logs payments as they already do in our USSD menu.
3. We match	Same place, same amount, within 24 hours. If the payer gives an M-Pesa code, we match on that instead.
4a. Match	The payer gets an SMS receipt. For lodges, where guests need a reason to bother, they also get KES 10 airtime.
4b. Mismatch	The owner gets an SMS straight away, with the payer’s number so they can follow up.
5. Owner checks in	The owner dials in any time: confirmed today, unmatched, and money at risk.

What it looks like on the phone

USSD 1

```
Thibitisha
1. Confirm a payment
2. My receipts
```

USSD 2

```
Enter the place code
(on the notice board,
or ask the caretaker):
4471
```

USSD 3

```
Amount paid (KES):
8000
```

USSD 4

```
M-Pesa code, or 0
if you paid cash:
0
```

USSD 5

```
Confirm KES 8,000 to
Mwangaza Apts?
Your number will be
shared with the owner.
1. Yes 2. Cancel
```

USSD 6 (end)

```
Thank you. We will SMS
your receipt once the
caretaker's record
matches.
```

SMS to Akinyi (match)

```
Receipt: KES 8,000 rent,
flat B4, Mwangaza Apts,
28 Oct. Confirmed by
caretaker.
Ref TB-4471-0193
```

SMS to Wanjiru (mismatch)

```
Alert: tenant
0712***345 (B4) says
they paid KES 8,000 on
28 Oct. Caretaker
logged KES 6,000.
```

Owner menu

```
Mwangaza Apts, Oct
Confirmed: 11
Unmatched: 1
At risk: KES 2,000
```

The story again, with Thibitisha

Wanjiru puts a notice on the building board: *"Paid rent? Dial the code and confirm. You will get a receipt."* Akinyi pays her KES 8,000 and confirms in under a minute. If Otieno logs it, Akinyi gets her receipt and the matter is closed. If he logs KES 6,000, or nothing, Wanjiru's phone tells her that same day, in Doha, with Akinyi's number.

And if Otieno is honest, every month shows **all matched**. For the first time he can prove it. That matters when we sell: we are not selling suspicion; we are selling receipts for everyone.

4. Why we believe it works

Evidence	What it shows
Brazil, Nota Fiscal Paulista (from 2007). Study by Joana Naritomi, American Economic Review.	Customers who asked for receipts got rebates and could check and complain online. Reported retail revenue rose at least 21% over four years. After a shop's first customer complaint it reported 7% more receipts.
Rwanda, receipt lottery (2015) and TENGAMARA (2026). IGC and J-PAL research.	Big prizes, rare draws and SMS entry with three codes failed: only 4% of receipts were submitted. People said they want small, frequent, certain rewards.
Kenya, KRA "Jinice na Risiti Legit" (2024).	Kenya's tax office tried the same idea, but only through a smartphone app, and only for seven weeks.
Kenya Communications Authority (June 2026).	27.4 million feature-phone devices that no receipt app reaches. Every one of them can dial USSD.

What we took from it

- Keep it to one short USSD session. No codes to remember except the place code on the wall.
- If there is a reward, make it small and certain (airtime on every match), never a lottery. That also keeps us away from gambling rules.

- Start where the payer already wants a receipt, so we do not need to pay them at all.

5. Who it is for, in order

Who	Does the payer want a receipt?	Reward needed?	Why this order
1. Landlords with caretakers (rent)	Yes, to protect themselves in disputes	No	Absent and diaspora owners feel the pain most and can pay
2. Chamas (contributions)	Yes, members distrust treasurers	No	The group itself is the customer
3. Milk collection centres	Yes, farmers dispute litres and pay-outs	No	Cooperatives already keep records to check against
4. Lodges and bars (guests)	No	Yes, KES 5-10 airtime	Our original customers; reuses the Shadow POS

All four use the same engine. Only the wording, the reward setting and who counts as “staff” change. That is why it suits the Africa’s Talking Marketplace: one product, installed separately by each business.

6. How we make money

	Rent	Lodges
Who pays	The landlord	The owner
Price	KES 50 per unit per month (a 20-flat block pays KES 1,000)	Our existing plans, KES 50-200 a day, with airtime passed on at cost
Our cost per confirmation	About KES 2.50 (USSD session ~1.50, SMS ~1)	About KES 12.50 (including KES 10 airtime)
Margin	About 94%	Set by the plan; rewards cost us nothing
Break-even	About 250 units (e.g. 12 blocks of 20) covers the KES 10,000/month USSD code and hosting	About 5 paying venues

Marketplace plans: Starter KES 1,500/month (up to 30 units or rooms), Standard KES 3,500/month (up to 100). Costs come from VALIDATION.md and UNIT_ECONOMICS.md; all prices are assumptions to test in the pilot.

Why it can grow

Startups that last in Africa own a trusted record of money moving every day and take a small cut of it (M-KOPA, Moniepoint, Wave). Startups that died owned trucks and stock (Twiga, Copia) or kept free records nobody trusted (Kippa). A payment confirmed by both sides is a trusted record. Later, a tenant’s confirmed rent history could become a rent reference or help them get credit. That is a later step, not today’s pitch.

7. What we have built so far

All code is in **/mnt/project-files/ussd-pos** (Node.js, Express, PostgreSQL/Supabase, Africa’s Talking). It was built for the lodge version and most of it carries over.

Install dependencies first (`npm install`), then `npm test`. Without that step the tests fail with “Cannot find module ‘pg’”.

Piece	What it does	Status	Reused for Thibitisha?
USSD engine (router)	Reads Africa’s Talking requests, walks menus, adds a confirm screen before every save, keeps screens under 182 characters, answers within time limits.	Done, tested	Yes, as is
Role menus by phone number	Manager, housekeeper and owner each see their own menu; unknown numbers are refused.	Done, tested	Yes; add “payer”

Piece	What it does	Status	Reused for Thibitisha?
Staff logging	Open and close shifts, log room cash and bar tabs, shortage or overage at close.	Done, tested	Yes, the “staff log” we match against
Owner SMS alerts	Alerts on shift close, on housekeeping mismatches and (on some plans) on every sale. Sandbox or live.	Done, tested	Yes, for mismatch alerts
Housekeeping cross-check	Cleaner’s room count against rooms logged.	Done, tested	Yes, for lodges
Owner menu	Cash held now; today’s takings.	Done, tested	Extend with confirmed/unmatched
Database	Supabase schema with append-only payments, row-level security, demo data; memory store for demos.	Done, tested	Yes; add payer tables
Tests	13 tests on the demo store (re-run 29 Oct: all pass); 25 including PostgreSQL on 14 Oct.	Passing	Add tests for matching
Research and pricing	VALIDATION.md, UNIT_ECONOMICS.md, USER_JOURNEYS.md, three research PDFs.	Done	Yes

8. What is left

Today, at the hackathon

#	Task	Done when	Suggested owner
0	Ask the organisers whether we may start from code written before today.	We know whether to reuse or rebuild	Kevin
1	Payer tables and store functions (places, payers, claims, matches).	Tests pass on the demo store	Backend
2	Payer USSD menu: place code, amount, M-Pesa code, consent, confirm.	A claim saves from the simulator	USSD
3	Matcher, SMS receipt on match, owner alert on mismatch.	Both SMS show in the sandbox	Backend
4	Airtime reward on a match (lodge template only).	Sandbox airtime call succeeds	Integrations
5	Owner menu: confirmed, unmatched, money at risk. Rent and lodge wording.	Owner sees live numbers	USSD
6	Dockerfile, environment settings, Marketplace listing, logo, screenshots.	Image builds; listing filled in	Product
7	Demo rehearsal with three simulator phones and three slides.	Demo runs in under 3 minutes	Everyone
Stretch	Owner texts “nani hajalipa?” (“who hasn’t paid?”) and an AI answers only from our records.	Correct answer on demo data	Integrations

After the hackathon

- **Talk to 10 landlords and 5 lodge owners** before writing more code. Ask how they check rent today and what a missing month costs them.
- **Pilot for four weeks** with 2 landlords and 2 lodges. Stop or change course if fewer than 30% of tenants confirm after two reminders, or if fewer than 2 of the 4 owners pay for month two.
- **Legal checks:** a lawyer’s view on airtime rewards under the Gambling Control Act 2025, and whether we must register with the Data Protection Commissioner (ODPC).

- **Go live:** buy a shared USSD code (KES 10,000/month, Safaricom only) only when a pilot is ready. The sandbox is free until then.
- **Pick the final name** and check it is free. The name lives in one setting, so changing it is easy.
- **Later:** Voice call-backs for disputes (needs a production account), a web dashboard, and rent references built on confirmed history.

9. Hard questions, honest answers

Question	Answer
“Why would a payer bother?”	Tenants and chama members want proof; that is why we start with them. Guests get airtime. If payers do not confirm in the pilot, we stop.
“Why not just use my till?”	Keep using it. We cover the money the till cannot see: cash and payments to staff numbers.
“Can staff bribe payers not to confirm?”	Sometimes; Rwanda saw shops offer discounts to skip receipts. Few tenants trade proof of rent for a small discount. For lodges we keep the cleaner check.
“Can someone send false claims?”	Rewards are paid only on a match, so there is nothing to farm. Every claim carries the caller’s number, rent places only accept registered tenants, and claims per number are limited.
“Is this legal?”	Payers agree on screen before we share their number. We keep only phone, amount and time. No lotteries. The two legal checks above are still open.
“Will investors care?”	Maybe later. Kenya raised only $46M of startup equity in the first half of 2026, so we plan to earn from customers first.

Words we use

Word	Meaning
USSD	The *123# style menus that work on every phone, with no internet or app.
Africa’s Talking (AT)	The Kenyan company whose USSD, SMS and Airtime services we build on. Its sandbox is a free test version.
AT Marketplace	AT’s store where businesses install apps built on AT. Each business gets its own copy.
Match / mismatch	Whether the payer’s confirmation agrees with what staff logged.
Chama	A savings group whose members contribute regularly to a shared pot.

Further reading in the project folder: research/Hackathon-What-To-Build-2026-10-29.pdf (how we chose this, sources and scoring), research/African-Startup-Trends-2026.pdf, ussd-pos/Informal-Economy-Research.pdf, ussd-pos/README.md.